

Math 241, F04
Exam 3

Name:
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	20	
2	20	
3	30	
4	15	
5	15	
Total	100	

Good Luck !

Problem 1. (20 points)

a) Evaluate $\int_1^{\frac{1}{2}} \int_0^{2y} \cos(\pi y^2) \, dx \, dy$.

Solution:

$$\begin{aligned} \int_1^{\frac{1}{2}} \int_0^{2y} \cos(\pi y^2) \, dx \, dy &= \int_1^{\frac{1}{2}} [x \cos(\pi y^2)]_0^{2y} \, dy \\ &= \int_1^{\frac{1}{2}} 2y \cos(\pi y^2) \, dy \\ &= \left[\frac{1}{\pi} \sin(\pi y^2) \right]_1^{\frac{1}{2}} \\ &= \frac{1}{\pi} (\sin(\pi/4) - \sin(\pi)) \\ &= \frac{1}{\pi} \sin(\pi/4) \end{aligned}$$

b) Evaluate $\int_0^2 \int_1^z \int_0^{\sqrt{x/z}} 2xyz \, dy \, dx \, dz$.

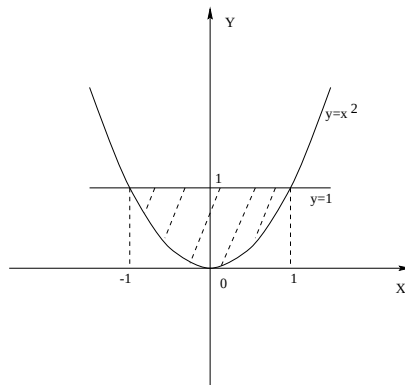
Solution:

$$\begin{aligned} \int_0^2 \int_1^z \int_0^{\sqrt{x/z}} 2xyz \, dy \, dx \, dz &= \int_0^2 \int_1^z [xy^2 z]_0^{\sqrt{x/z}} \, dx \, dz \\ &= \int_0^2 \int_1^z x^2 \, dx \, dz \\ &= \int_0^2 \left[\frac{x^3}{3} \right]_1^z \, dz \\ &= \int_0^2 \frac{1}{3} (z^3 - 1) \, dz \\ &= \frac{1}{3} \left[\frac{z^4}{4} - z \right]_0^2 \\ &= \frac{1}{3} (4 - 2) = \frac{2}{3} \end{aligned}$$

Problem 2. (20 points) Let S be the region bounded by $y = x^2$ and $y = 1$ in the plane.

a) Sketch the region S and write it down in the form of y -simple set. (*Remember to label the curves and the axes.*)

Solution:



$$S = \{(x, y) \mid -1 \leq x \leq 1, x^2 \leq y \leq 1\}$$

b) Evaluate $\iint_S xy \, dA$.

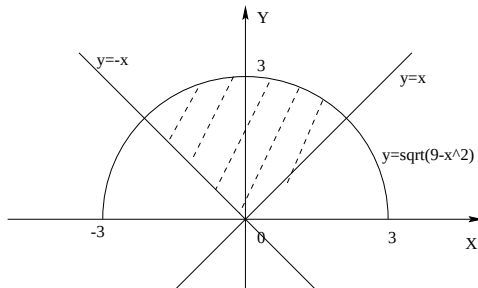
Solution:

$$\begin{aligned} \iint_S xy \, dA &= \int_{-1}^1 \int_{x^2}^1 xy \, dy dx \\ &= \int_{-1}^1 \left[\frac{xy^2}{2} \right]_{x^2}^1 dx \\ &= \int_{-1}^1 \frac{1}{2}(x - x^5) \, dx \\ &= \frac{1}{2} \left[\frac{x^2}{2} - \frac{x^6}{6} \right]_{-1}^1 \\ &= 0 \end{aligned}$$

Problem 3. (30 points) A lamina covering the region S in the plane bounded by the curves $y = x$, $y = -x$ and $y = \sqrt{9 - x^2}$ is associated with a density $\delta(x, y) = \sqrt{x^2 + y^2}$.

a) Sketch S and write it down in the form of polar rectangle.

Solution:



$$S = \{(r, \theta) \mid 0 \leq r \leq 3, \frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4}\}$$

b) Find the mass of the lamina m by evaluating $\iint_S \delta(x, y) dA$ in polar coordinates.

Solution:

$$\begin{aligned} m &= \iint_S \sqrt{x^2 + y^2} dA = \int_{\pi/4}^{3\pi/4} \int_0^3 r \cdot r dr d\theta \\ &= \int_{\pi/4}^{3\pi/4} \left[\frac{r^3}{3} \right]_0^3 d\theta = \int_{\pi/4}^{3\pi/4} 9 d\theta \\ &= 9 \cdot (3\pi/4 - \pi/4) = \frac{9\pi}{2} \end{aligned}$$

c) Find the center of the mass $(\bar{x}, \bar{y})$ of the lamina.

Solution: First by the symmetry of S and $\delta(x, y)$ with respect to y -axis, we have $\bar{x} = 0$.

$$\begin{aligned} M_x &= \iint_S y \sqrt{x^2 + y^2} dA = \int_{\pi/4}^{3\pi/4} \int_0^3 r^2 \sin \theta \cdot r dr d\theta \\ &= \int_{\pi/4}^{3\pi/4} \sin \theta \left[\frac{r^4}{4} \right]_0^3 d\theta = \int_{\pi/4}^{3\pi/4} \frac{81}{4} \sin \theta d\theta \\ &= \frac{81}{4} [-\cos \theta]_{\pi/4}^{3\pi/4} = \frac{81}{2\sqrt{2}} \end{aligned}$$

$$\text{So we have } \bar{y} = \frac{M_y}{m} = \frac{81}{2\sqrt{2}} \cdot \frac{2}{9\pi} = \frac{9\sqrt{2}}{2\pi}.$$

Problem 4. (15 points) Let G be the part of plane $3x + 4y + 6z = 12$ that is above the rectangle R in the xy -plane with vertices $(0,0)$, $(2,0)$, $(2,1)$, and $(0,1)$. Find the area of G . (*Hint:* let $f(x, y) = \frac{1}{6}(12 - 3x - 4y)$.)

Solution: Let

$$f(x, y) = \frac{1}{6}(12 - 3x - 4y) = 2 - \frac{1}{2}x - \frac{2}{3}y,$$

then

$$f_x = -\frac{1}{2}, \quad f_y = -\frac{2}{3}.$$

We also have

$$R = \{(x, y) \mid 0 \leq x \leq 2; 0 \leq y \leq 1\}.$$

So the area of G is given by

$$\begin{aligned} A(G) &= \iint_R \sqrt{f_x^2 + f_y^2 + 1} \, dA \\ &= \int_0^2 \int_0^1 \sqrt{(-1/2)^2 + (-2/3)^2 + 1} \, dydx \\ &= \int_0^2 \int_0^1 \frac{\sqrt{61}}{6} \, dydx \\ &= \frac{\sqrt{61}}{6} \cdot 2 \\ &= \frac{\sqrt{61}}{3} \end{aligned}$$

Problem 5. (15 points) Let S be the solid bounded below by the paraboloid $z = x^2 + y^2$ and above by the plane $z = 9$. The iterated triple integral in Cartesian coordinates for the volume of S is

$$V(S) = \int_0^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_{x^2+y^2}^9 dz \, dy \, dx$$

a) Convert the triple integral for the volume of S to cylindrical coordinates.

Solution:

$$S = \{(r, \theta, z) \mid 0 \leq r \leq 3, 0 \leq \theta \leq 2\pi, r^2 \leq z \leq 9\}$$

$$V(S) = \int_0^{2\pi} \int_0^3 \int_{r^2}^9 r \, dz \, dr \, d\theta$$

b) Evaluate one of the two triple integrals to find the volume of S .

Solution: Let us use the triple integral in cylindrical coordinates.

$$\begin{aligned} V(S) &= \int_0^{2\pi} \int_0^3 r [z]_{r^2}^9 \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^3 r(9 - r^2) \, dr \, d\theta \\ &= \int_0^{2\pi} \left[\frac{9r^2}{2} - \frac{r^4}{4} \right]_0^3 \, d\theta \\ &= \int_0^{2\pi} \left(\frac{81}{2} - \frac{81}{4} \right) \, d\theta \\ &= \frac{81}{4} \cdot 2\pi \\ &= \frac{81\pi}{2} \end{aligned}$$